ICE4405P
Digital Signal Processing

Spring Semester 2026

Course Information

Instructor:	Longguang Li (llg9012@sjtu.edu.cn)
Textbooks:	Alan V. Oppenheim and Ronald V. Schafer, *Discrete-Time Signal Processing*, third edition, Prentice–Hall.
Notes	Course slides are posted on the course Web Site as PDF files.
Modeling Software:	Matlab (Recommanded), Mathematica, others
Office Hours:	Monday after class, others by appointment

Course Information

Grading:

1. Attendance 10%
2. Graded homework assgninment 30%
3. Lab (3 Labs) 20%
4. Final exam 40%.

Topics and Text Sections

Topics	Text Sections
1. Introduction/Course Overview	
2. Discrete-time signals and systems	2.0–2.9
3. Sampling of continuous-time signals	4.0–4.6
4. The z-transform	3.0–3.4
5. Transform analysis of linear time-invariant systems	5.0–5.7
6. The discrete Fourier transform	8.0–8.7
7. Computation of the discrete Fourier transform	9.0–9.6
8. FIR/IIR filters and random signal analysis	7.0–7.5

Course Objective

The expected learning outcomes of this course are a more in-depth treatment of discrete-time signals and systems. As a discipline within electrical engineering this known as digital signal processing (DSP).
The students will learn

- model discrete-time signals and systems in the time domain;
- extend the time domain modeling to the frequency domain using the discrete- time Fourier transform (DTFT);
- working signals and linear time invariant (LTI) systems using z-transform (ZT) techniques;
- sampling theory and multirate sampling theory as found in modern DSP;
- DSP problem solving using time, frequency, and z-domains effectively;
- the value and power of the discrete Fourier transform (DFT) and its efficient implementation via fast Fourier transform (FFT) algorithms;
- simulation of DSP algorithms and subsystems using Matlab.

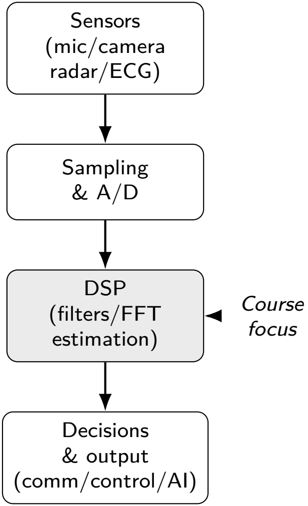
Background

- The world is *analog* (sound, light, motion, voltage), but computers are *discrete*.
- DSP provides the core ideas for turning raw measurements into *usable information*: filtering, spectral analysis, estimation, compression, and detection.
- If a system *senses*, *communicates*, or *makes decisions*, there is almost always a DSP layer somewhere inside.

DSP is the mathematical glue between sensors and intelligent action.

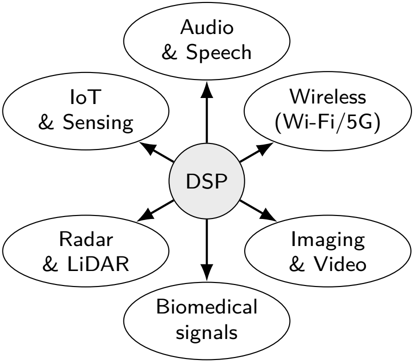
DSP is the universal signal chain

- Your microphone, camera, ECG electrode, radar antenna, and accelerometer all output *signals*.
- DSP is what lets us:
  - remove noise/interference,
  - estimate hidden quantities (frequency, delay, direction),
  - compress for storage/streaming,
  - detect events reliably.
- Learning DSP also teaches a transferable way of thinking: time domain ↔ frequency domain ↔ z-domain.



DSP applications today

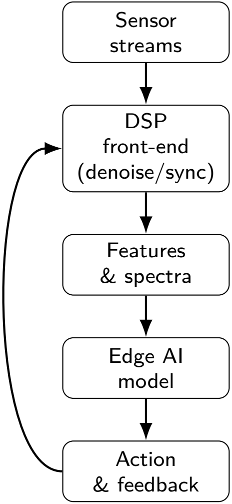
- **Audio & speech:** noise cancellation, echo suppression, beamforming, codecs.
- **Wireless:** modulation/demodulation, channel equalization, OFDM, MIMO.
- **Imaging & video:** enhancement, denoising, compression, stabilization.
- **Sensing:** radar/sonar/LiDAR processing, localization, tracking.
- **Biomedical:** ECG/EEG filtering, wearable monitoring, medical imaging pipelines.



DSP applications tomorrow

- **6G / joint communication & sensing:** waveforms that both *transmit data* and *measure the environment*.
- **Autonomous systems:** sensor fusion with strict latency and reliability constraints.
- **AR/VR & spatial computing:** low-latency audio/video pipelines (your brain is picky).
- **Edge AI:** DSP front-ends that make machine learning robust, efficient, and deployable.
- **New sensors:** ultrasound arrays, hyperspectral imaging, brain–computer interfaces.

DSP + ML = modern pipelines



Course topics $\longrightarrow$ practical DSP skills

Course topic	**What it lets you build / debug**
Sampling & quantization	Reliable A/D interfaces; anti-aliasing; understanding limits of sensors and hardware.
LTI systems & filtering	Noise removal, interference suppression, equalization, feature extraction.
DTFT & Z-transform	Frequency response intuition; stability/causality; efficient system design.
DFT / FFT	Fast spectrum analysis; fast convolution; OFDM; spectral features for ML.

If you can think in time, frequency, and z domains, you can design systems that work *and* explain why they work.